

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

Semester: January – May 2021

In-Semester Examination**Class: SY B. Tech****Branch: Computer Engineering****Semester : IV****Full name of the Course: Relational Database Management Systems****Course Code: 2UCC403****Duration: 1hr.15 min (attempting questions) +15 min (uploading) Max. Marks: 30**

Q. No	Questions	Marks
Q1	lives(<u>person-name</u>, street, city) works(<u>person-name</u>, company-name, salary) located-in(<u>company-name</u>, city) manages(<u>person-name</u>, manager-name) For the above schema (the primary key for each relation is denoted by the underlined attribute), provide relational algebra expressions for the following queries: Q1 A Find all tuples in works of all persons who work for the State Bank of India Company. Q1B Find the name of persons working at Bank of Baroda who earn more than ₹ 50,000. Q1C Find the name and city of all persons who work for Punjab National Bank and earn more than ₹50,231 Q1D Find names of all persons who live in the same city as the company they work for.	1 Marks each

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	lives(<u>person-name</u>, street, city) works(<u>person-name</u>, company-name, salary) located-in(<u>company-name</u>, city) manages(<u>person-name</u>, manager-name) For the above schema (the primary key for each relation is denoted by the underlined attribute), provide SQL Statement for the following queries: Q1 E Find names of all persons who live in the same city and on the same street as their manager. Q1 F Find names of all persons who do not work for Canara Bank. Q1 G Find the name of all persons who work for Yes Bank and live in Mathura.	2 Mark s each
Q2 A	What is an attribute? Explain different types of attributes with example (1 Mark for Definition of attribute, 1 mark for writing each attribute with its meaning and example) OR Define Armstrong Axioms used in Relational Database	05 marks

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	Design(1 mark for definition for Armstrong and listing with its meaning)	
Q2 B	What is referential integrity? Explain with suitable example. (Meaning of referential integrity for 1 mark, example for 2 mark and its explanation for 2 mark) OR What is Views in SQL(1 mark) How it is defined? (Syntax for 2 marks) Example (2 Marks)	05 marks
Q3 A	What is ER diagram? Give example of ER diagram.(1 mark for ER diagram meaning, and Example carry 4 marks)	5 Mark s
Q3 B	The overall design of database is called? A. Schema of the database B. Structure of the database C. The screen of database D. View of the database	1 Mark
Q 3 C	Example of non-numerical data is? A. Employee address B. Examination score C. Bank balance D. Employee Salary	1 Mark
Q 3 D	A data dictionary includes? A. Information about system data B. Procedure logic C. Project requirements	1 Mark

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	D. All of above	
Q 3 E	A table column that contains values that are primary-key values in another table? A. Primary key B. Relation Key C. Foreign key D. Super key	1 Mark
Q 3 F	The collection of related fields is called _____? A. Field B. File C. Record D. Both A and B	1 Mark